ROSE5730 Autonomous Mobile Robots

Lecture 1: Introduction

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Outline

Introduction

3 Examples

2 Key problems

1 Basic definitions



What is a Mobile Robot

- **Robot** (general): A machine capable of sensing, computation, and actuation to perform tasks.
- Mobile robot?
- **Mobile robot**: A robot with **locomotion** enabling it to move through its environment.



Locomotion

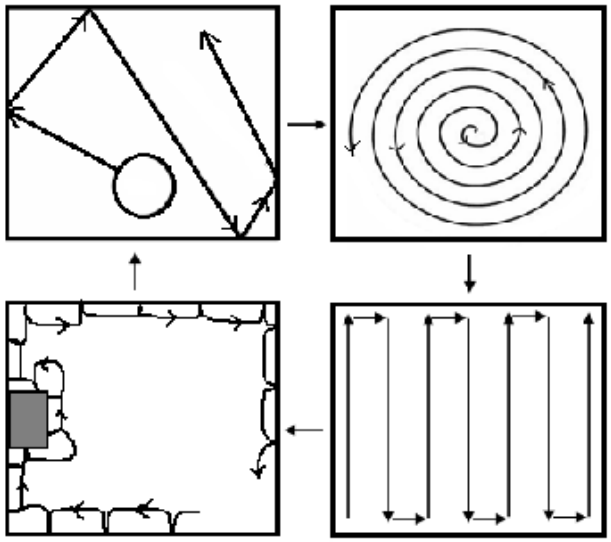
- **Locomotion** is the ability of a robot to convert actuation into controlled movement through its environment.





Autonomous Mobile Robot

Autonomous mobile robot: A mobile robot that perceives, decides, and acts with minimal human intervention using onboard sensing and computation.





Key distinctions

- Teleoperated vs Semi-autonomous vs **Autonomous**
- **Mobile** vs Fixed (industrial arm on a line)

Dimension	Teleoperated	Semi-autonomous	Autonomous
Control authority	**Human** fully in the loop; direct joystick/commands	**Shared control**; robot assists/executes sub-tasks	**Robot** primary authority; human supervises/sets mission
Perception & localization	Optional/basic (mainly for operator feedback)	Needed for assisted functions (e.g., obstacle avoidance, stabilization, docking)	Comprehensive onboard perception, localization, and **SLAM**
Planning	Human plans path and maneuvers	Robot plans **short-horizon** motions; human sets waypoints/goals	Robot plans **global and local** paths/trajectories under constraints
Execution & control	Human drives; robot may do low-level stabilization	Robot executes motion primitives (PID/model-based), handles routine hazards	Full stack control (planning → control → recovery behaviors)
Connectivity dependence	**High** (latency/links critical)	**Moderate** (can continue some functions if link drops)	**Low** (can operate offline; may report/seek help when needed)

Why Mobile Robots

- Societal & economic drivers:
  - Labor shortages, and safety in hazardous tasks
  - 24/7 operations, scalability, and operational efficiency



- Advantages over fixed automation:
  - Flexibility in dynamic environments
  - Lower infrastructure cost (compared to conveyors/ Automated Guided Vehicles with rigid infrastructure)
  - Rapid reconfiguration and software-defined behavior
  - Easy conversion from fixed-base robots

Why Mobile Robots

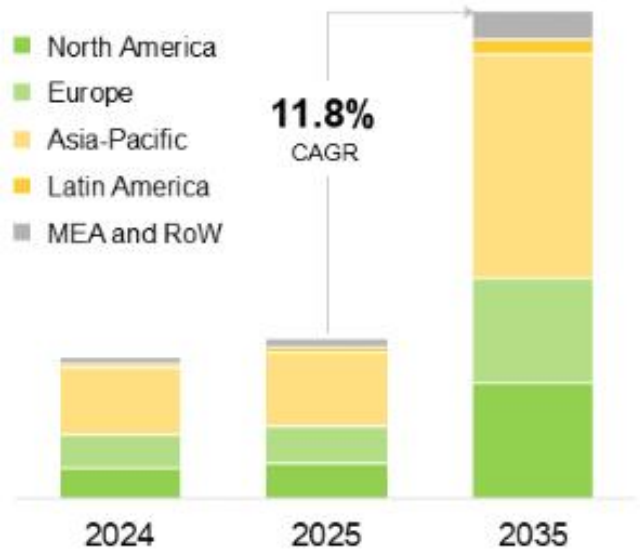
Mobile Robots Market

Roots Analysis
BUSINESS RESEARCH & CONSULTING

Market Distribution by Geographical Regions, Till 2035

Value in 2025
USD 20.79 Billion

Value in 2035
USD 63.28 Billion



Market Landscape

Distribution by Payload Support[1]







Distribution by Type of Metamaterial



Market Dynamics Insights

Competitive Landscape

With the presence of several small and large mobile robot manufacturing companies, the market is experiencing intense competition.

Emerging Spot

The market in Middle East and Africa, and Rest of the World is expected to grow at a relatively high CAGR (13.00%), during the forecast period till 2035.

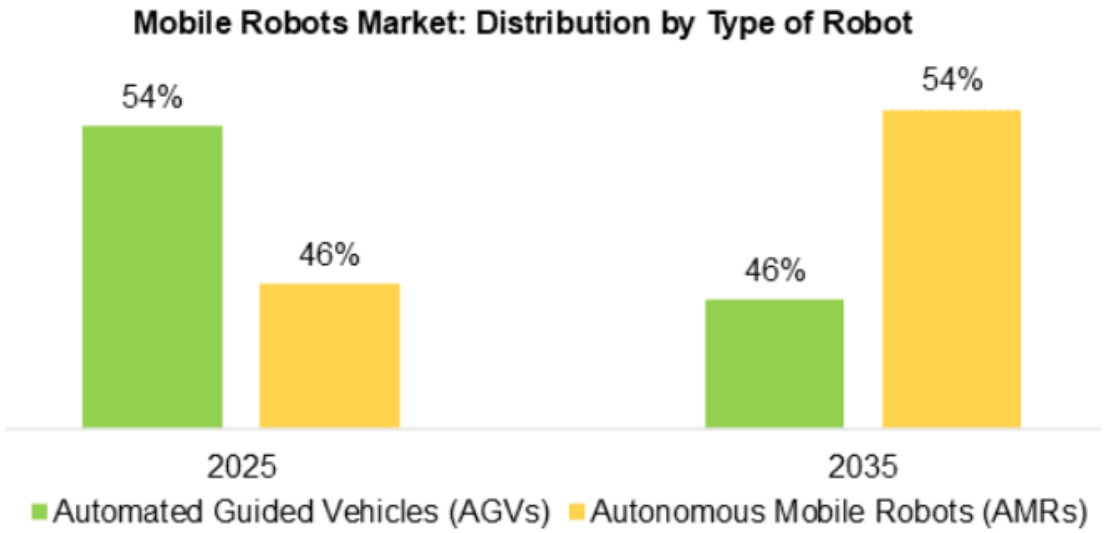
Trends

Impact of AI on mobile robots market, along with the rising adoption of healthcare mobile robots will play a crucial role in shaping the future of technological innovation.

Abbreviation: MEA and RoW: Middle East and Africa and Rest of the World
Note 1: Companies for which information on more than one payload support was available have been counted multiple times

Why Autonomous Mobile Robots?

- **Lower labor cost**: 24/7 operation, optimized routing and task allocation; no need for continuous operator attention (teleop) or frequent supervision (semi-autonomous).
- **Better adaptation and resilience**: Onboard perception/ localization/ planning (e.g., SLAM) lets AMRs re-plan around obstacles, recover from disruptions, and keep working even if connectivity drops.



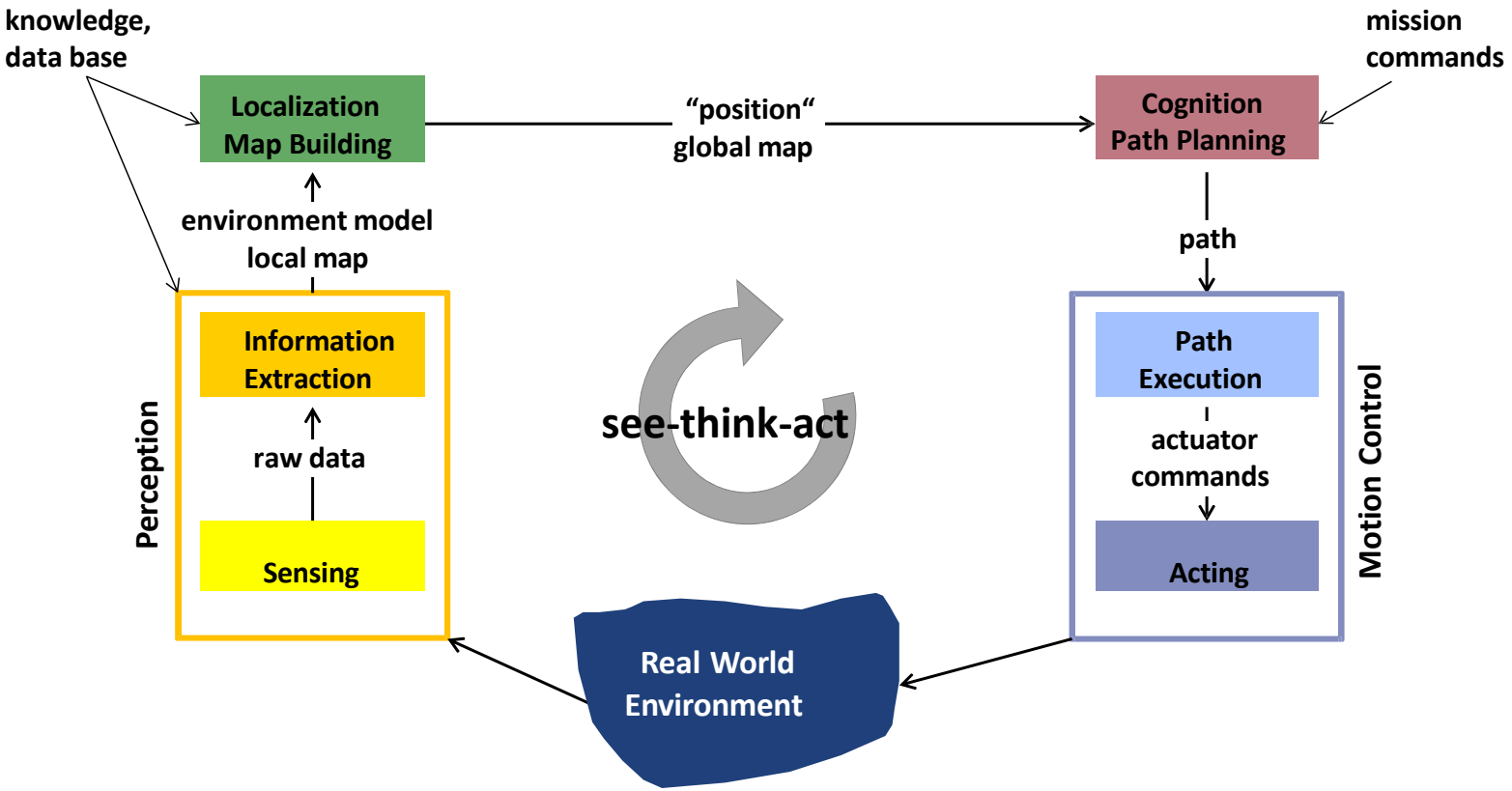
Key Problems

- The three key questions in Mobile Robotics
  - Where am I ?
  - Where am I going ?
  - How do I get there ?
- To answer these questions the robot has to
  - have a model of the environment (given or autonomously built)
  - perceive and analyze the environment
  - find its position/situation within the environment
  - plan and execute the movement

Key Problems

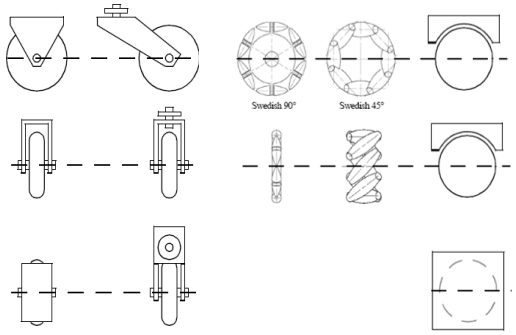
- **Locomotion and Kinematics**: Nonholonomic constraints (e.g., differential drive), maneuverability
- **Perception**: Robust sensing in varied lighting/weather; sensor fusion
- **Localization**: Estimating pose with uncertainty (odometry drift; GPS outages)
- **Mapping and SLAM**: Building consistent maps; loop closure; scalability
- **Navigation**: Generate safe and efficient paths; dynamic obstacle avoidance; multi-goal routing

See-think-act Cycle



kinematics and Motion Control

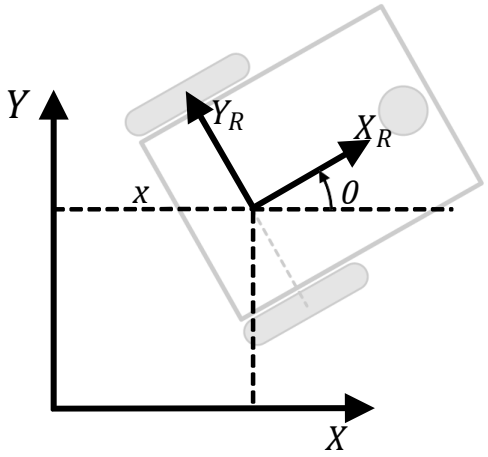
- Wheel types and its constraints
  - Rolling constraint
  - no-sliding constraint (lateral)



- Motion control

$$\begin{bmatrix} x' \\ y' \\ 0 \end{bmatrix} = f\left(i'_1 \cdots i'_n, 0, geometry\right)$$

$$\begin{bmatrix} i'_1 \\ \vdots \\ i'_n \end{bmatrix} = f\left(x, y, 0\right)$$

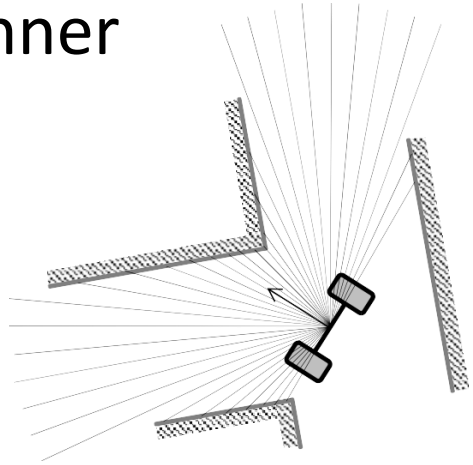


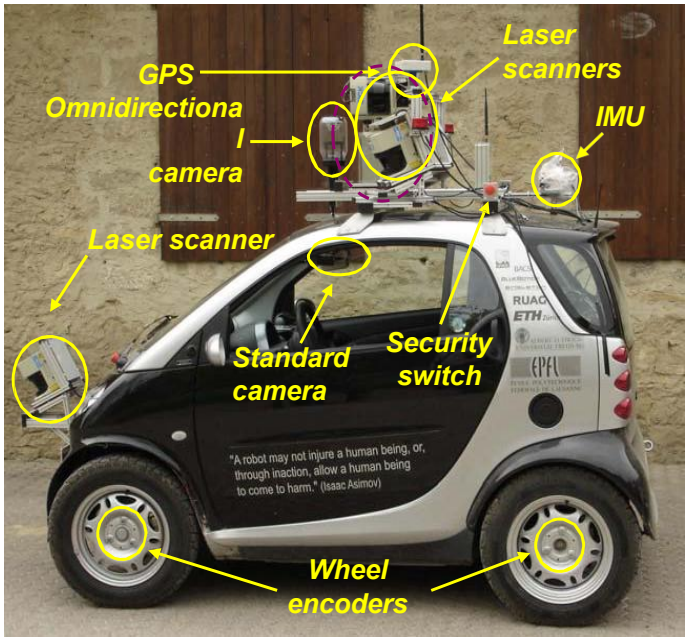


Perception

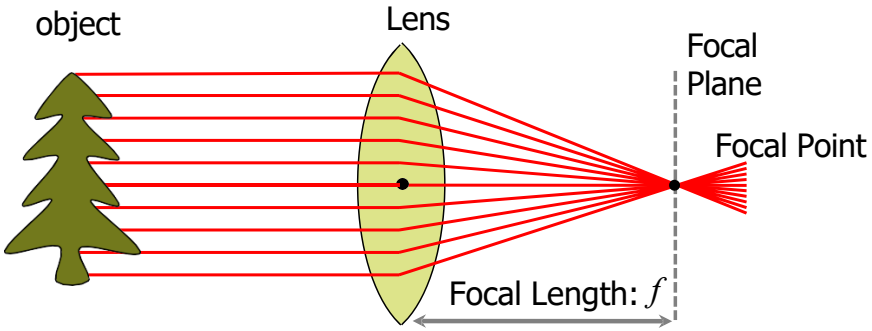
Laser scanner

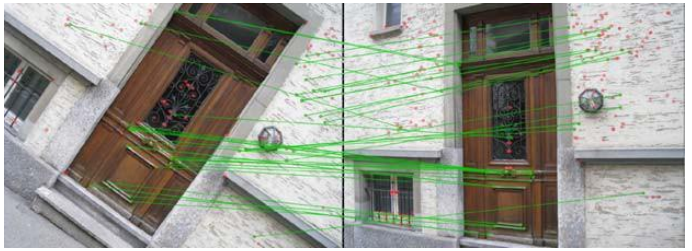




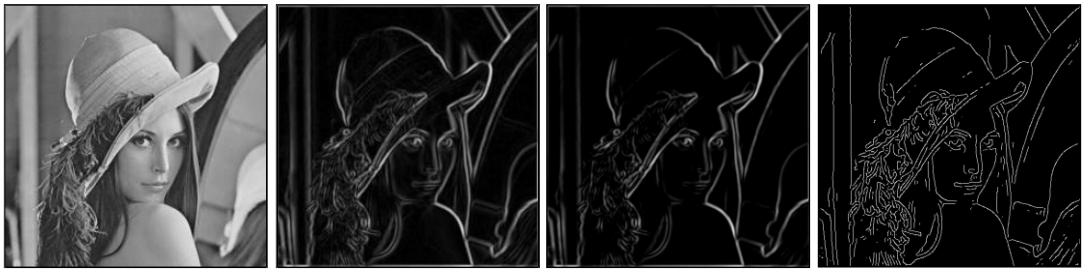


Camera

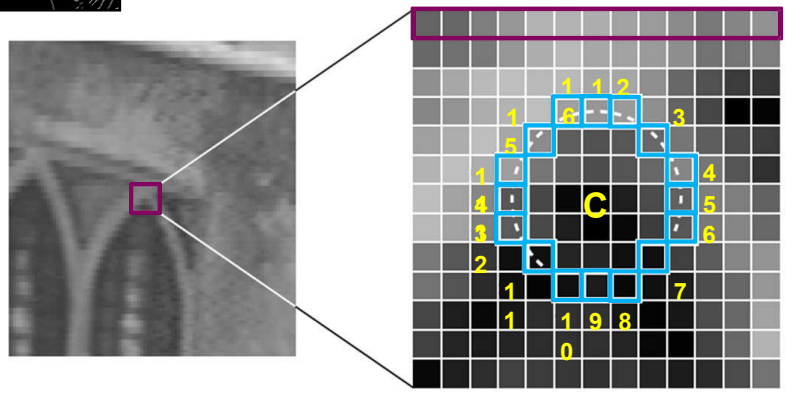




Perception



- Filtering / Edge Detection



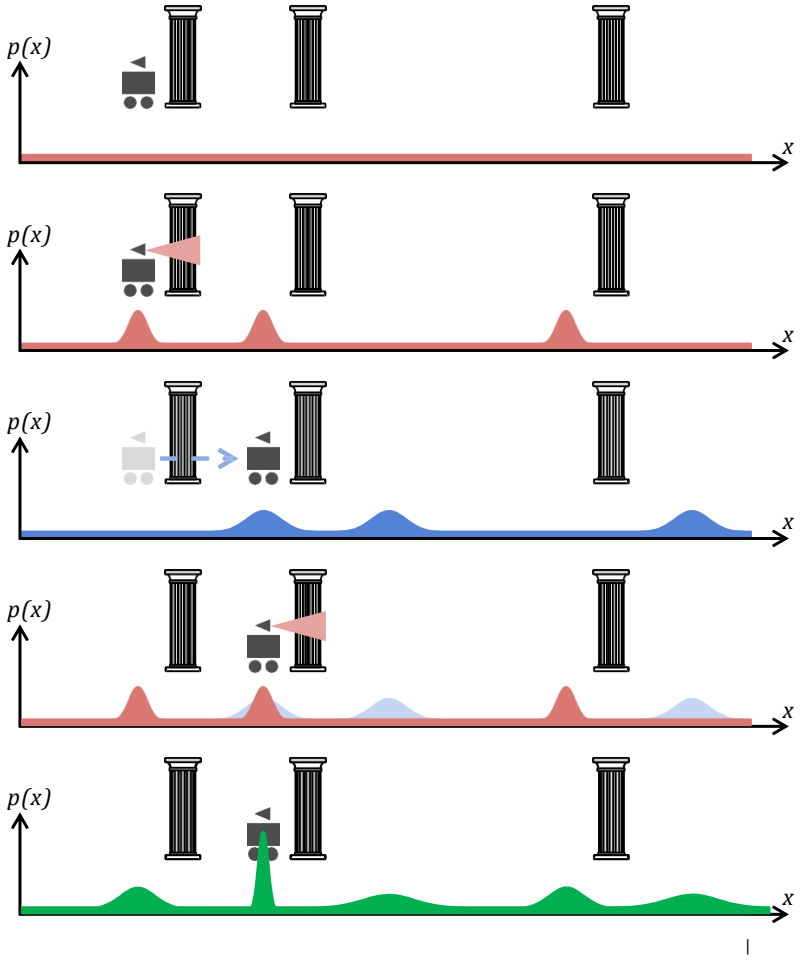
- Keypoint Features



- Keypoint matching, e.g., BRISK

Localization

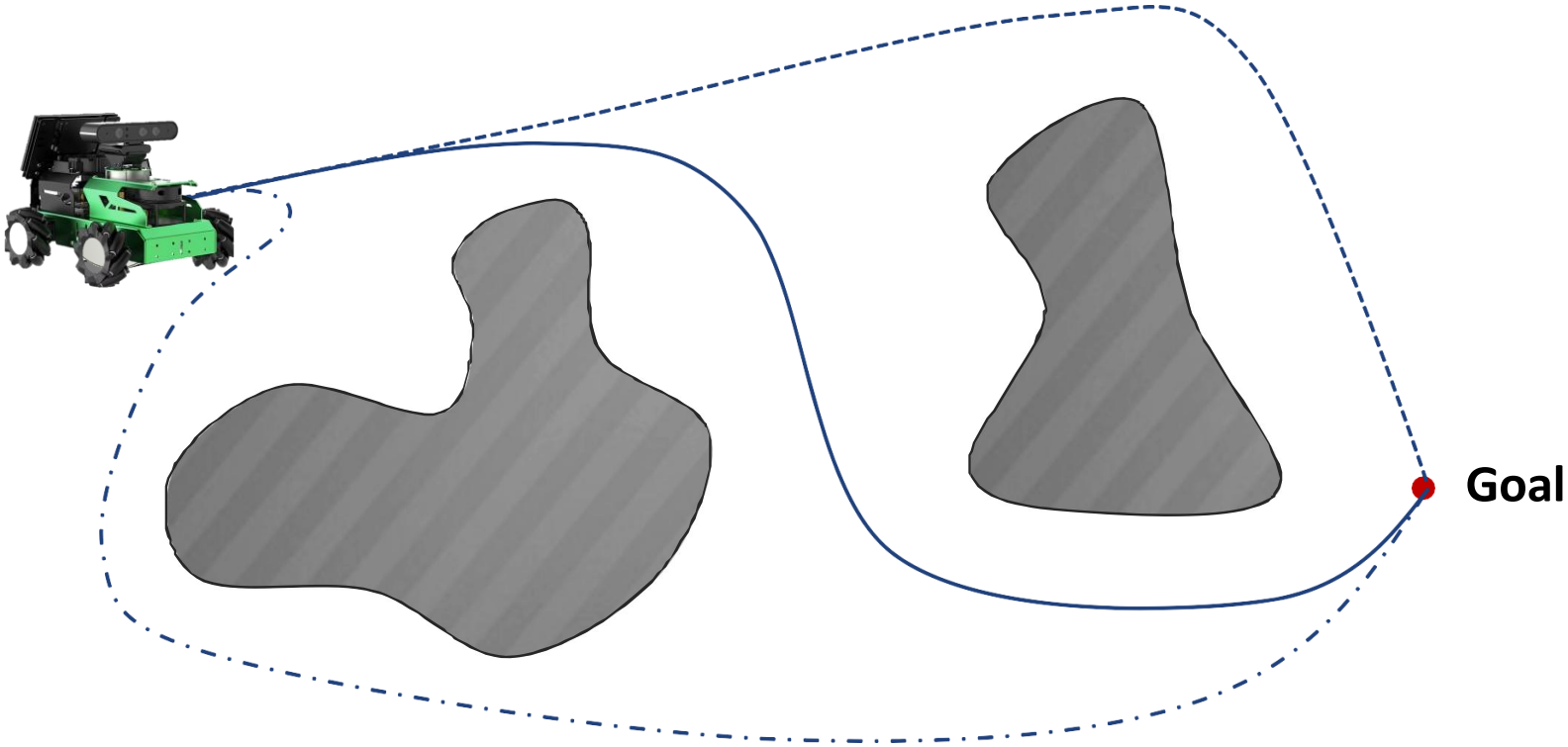
- **SEE**: The robot queries its sensors → finds itself next to a pillar
- **ACT**: Robot moves one meter forward
  - motion estimated by wheel encoders
  - accumulation of uncertainty
- **SEE**: The robot queries its sensors again → finds itself next to a pillar
- Belief update (information fusion)



Navigation

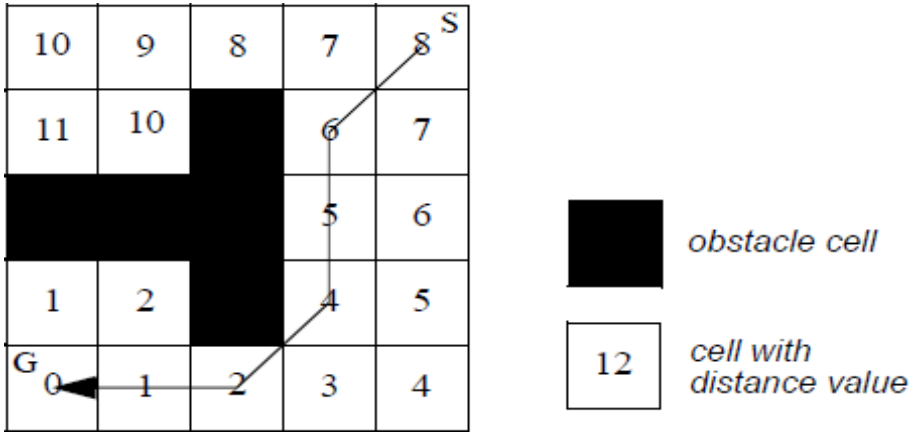
Where am I going ?

How do I get there ?

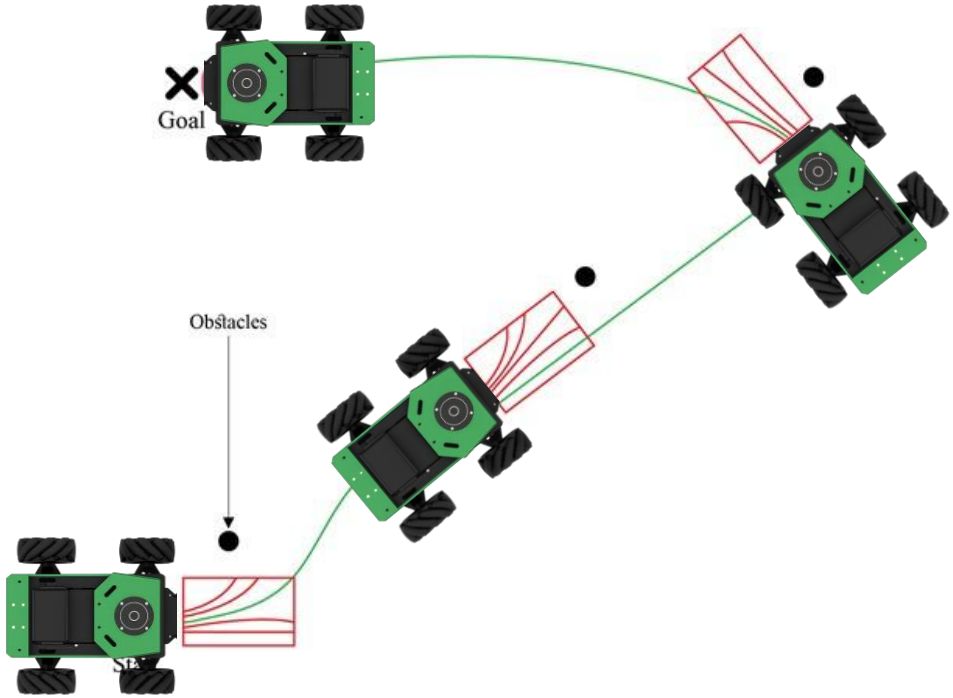


Navigation

- Global path planning: Graph search



- Local path planning: Collision avoidance



Main Categories by Locomotion

- **Wheeled**: differential, car-like, omnidirectional, tracked
- **Legged**: biped, quadruped, multi-legged
- **Aerial**: multirotor, fixed-wing, Vertical Take-Off and Landing (VTOL)
- **Aquatic**: Autonomous Underwater Vehicle (AUV), Remotely Operated Vehicle (ROV), surface vessels
- **Hybrid**: e.g., wheeled–legged, tilt-rotor







Robotic Teacher

Basic Development Procedures

C

Chatbot development

Textual conversational chatbot with basic training set of Arduino programming and robotic control.

T

Text to Speech Conversion

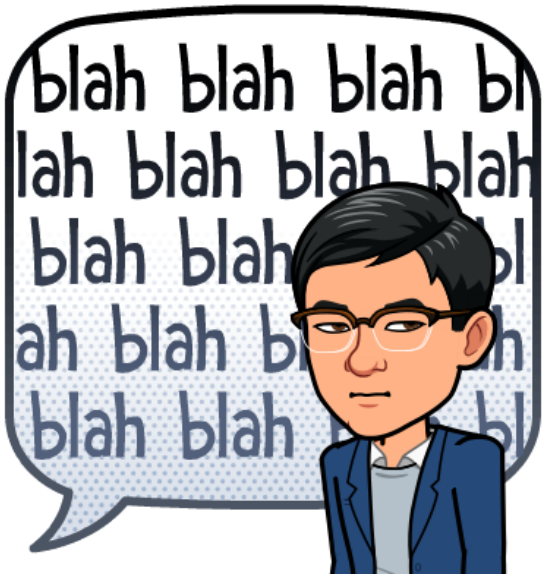
Clone the voice of teachers and generate a specific teacher's dialogue by using TTS technology.

P

Personal Learning APPs

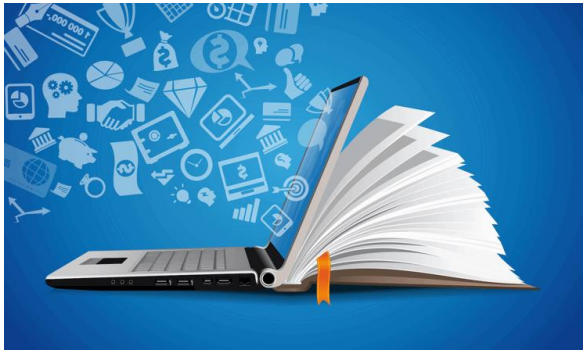
Individualized teaching material generation, face recognition, automatic assessment, and remote lab tour.

Voice Cloning









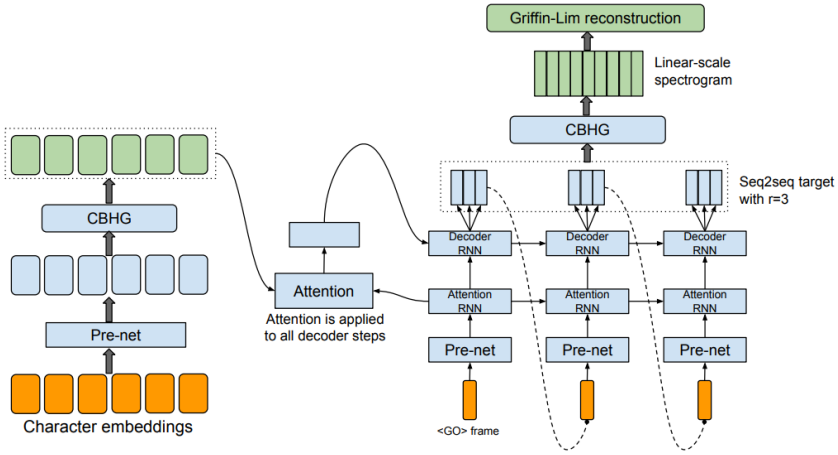
Statistical Parametric Synthesis

Four-step synthesis process

With a statistical parametric speech synthesis method

Synthesized voice





Vocoder

Convert Spectrogram or mel-spectrogram to waveform

W

c

Feature generation

The statistical model learns from speech corpora using machine learning techniques

d

Text analysis

Encode information of how speech evolves as a function of time in the context of a given input text.

A

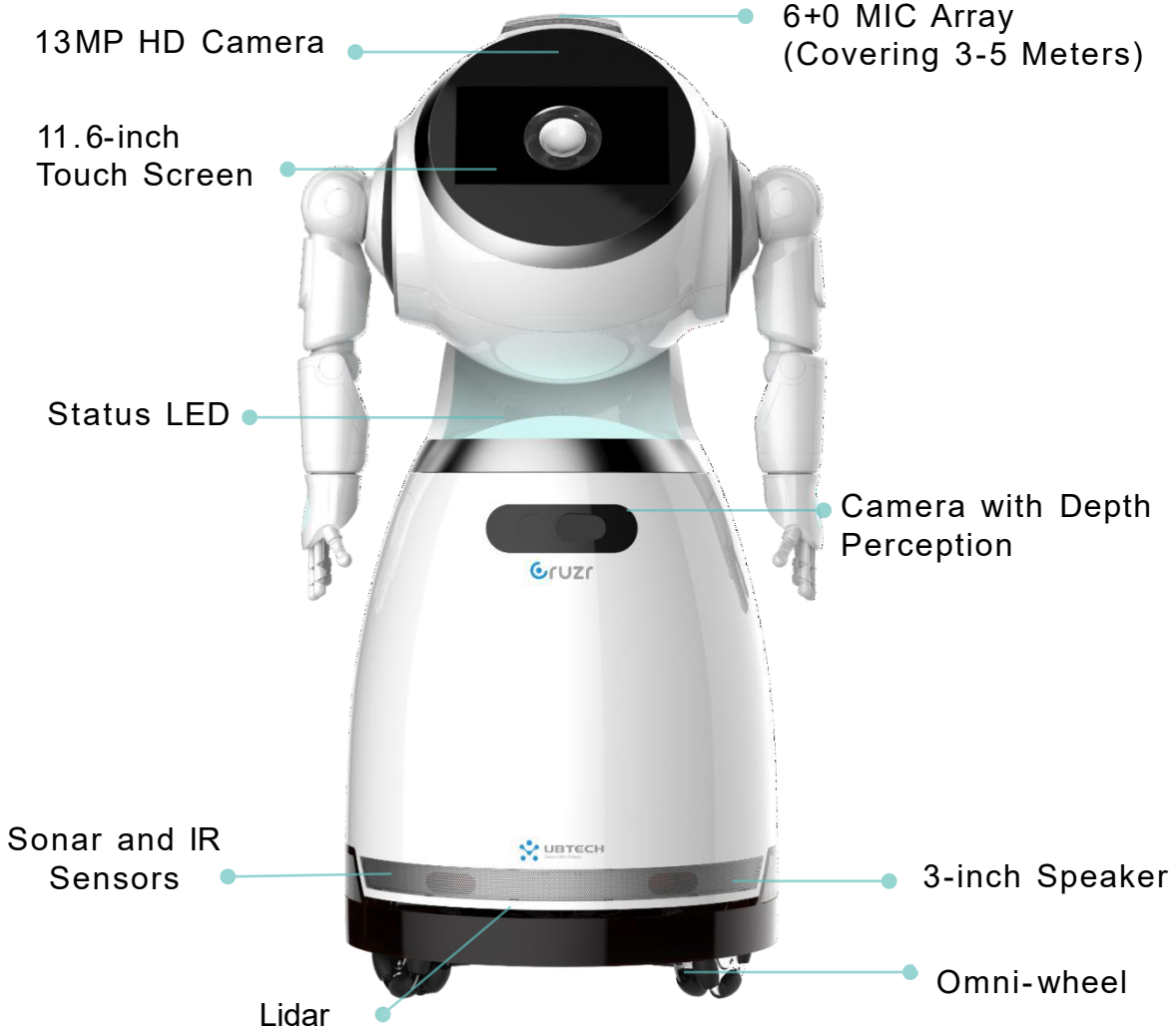
Text input

Provide the text to the system, e. g. “This is the text I want to synthesize”

Wheeled Mobile Robot: Case 1

Cruzr Robot

- Humanoid Design, Flexible Arms and 360° Movements
- With 17 servos, Cruzr moves like a human. With its Four-Layer Protection System Cruzr is capable of shaking hands, greeting visitors, moving around, dancing and hugging.



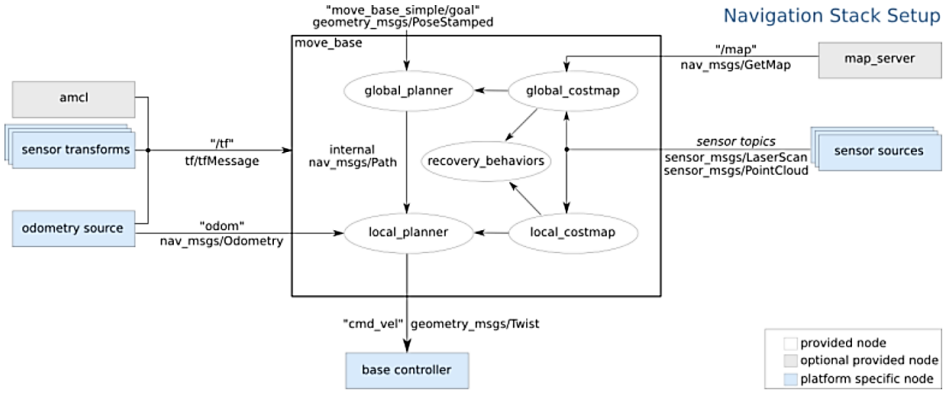
Wheeled Mobile Robot: Case 1

Slam-gmapping

Adaptive Monte Carlo Localization (AMCL) + odometry + laser scanning



Map of the lab AB1 206 CUHK from gmapping



Navigation stack setup on the robotic teacher

Navigation

- A* search algorithm
- Dynamic Window Approach (DWA)
- Model Predictive Control (MPC)

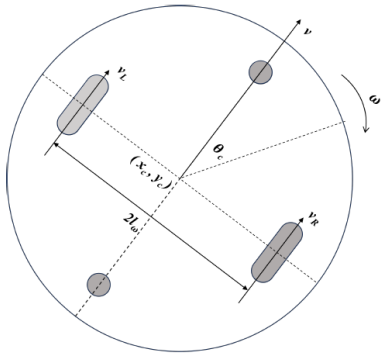
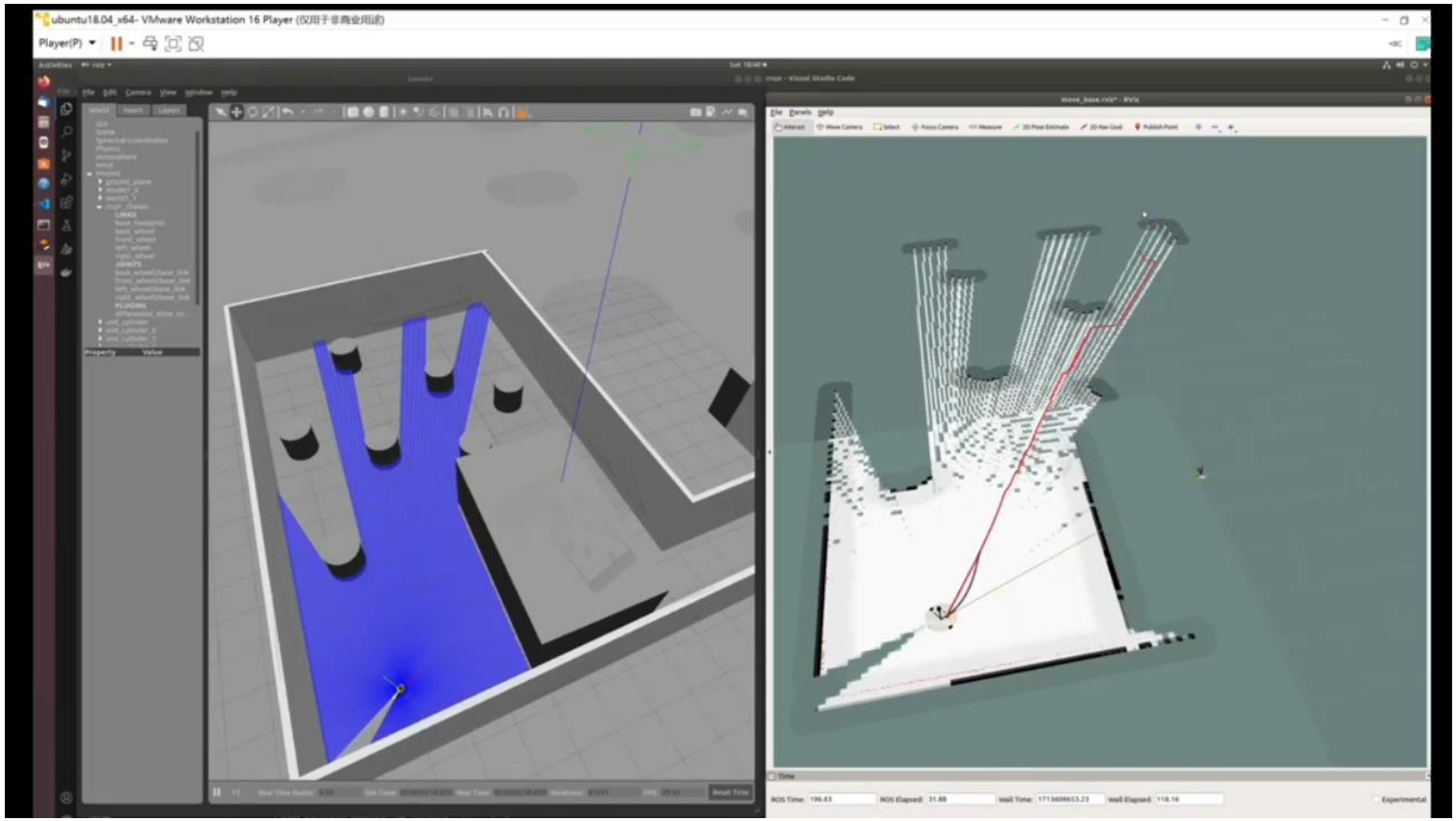


Illustration of robot chassis

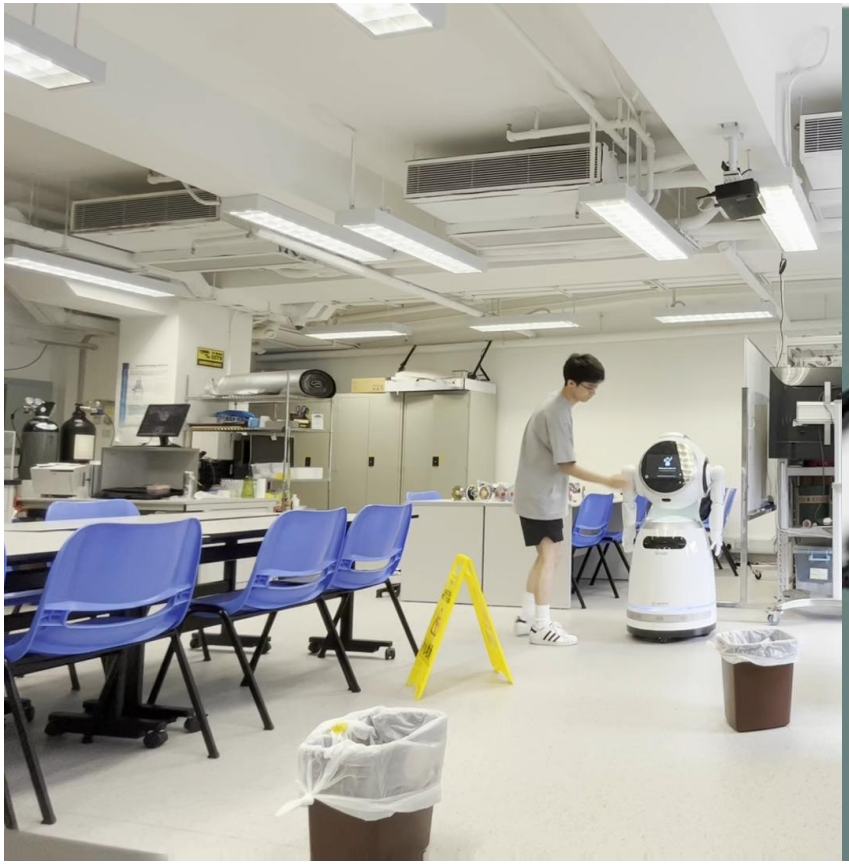
Wheeled Mobile Robot: Case 1

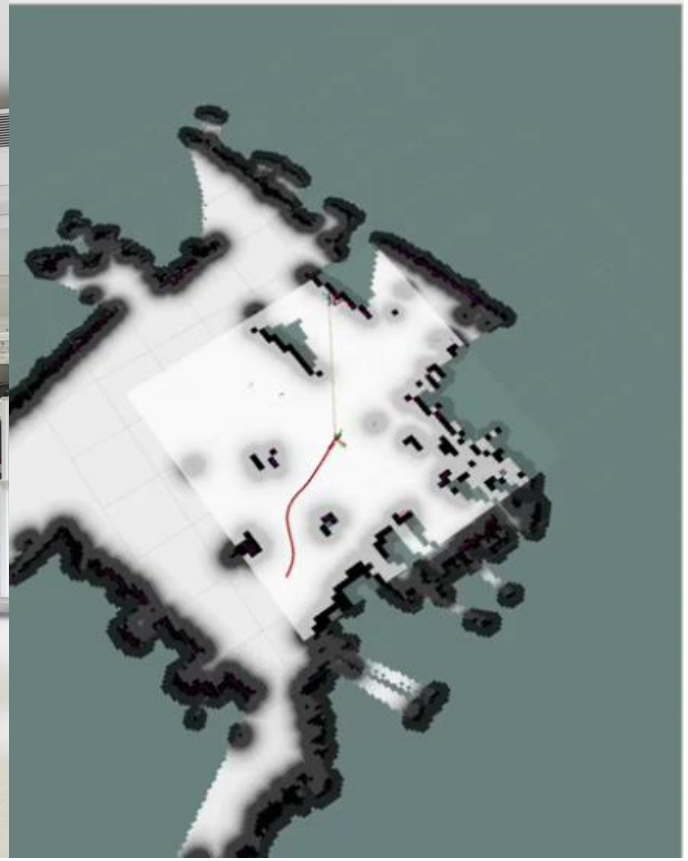
- **MPC path planner** in the simulator



Wheeled Mobile Robot: Case 1

Cruzr with DWA in the environment with a few obstacles

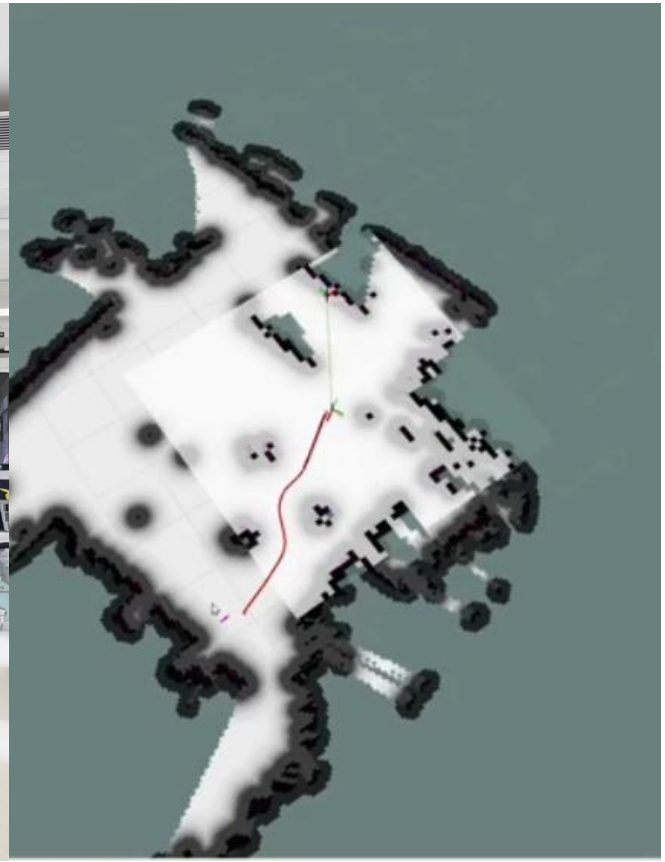




Wheeled Mobile Robot: Case 1

Cruzr with MPC in the environment with a few obstacles





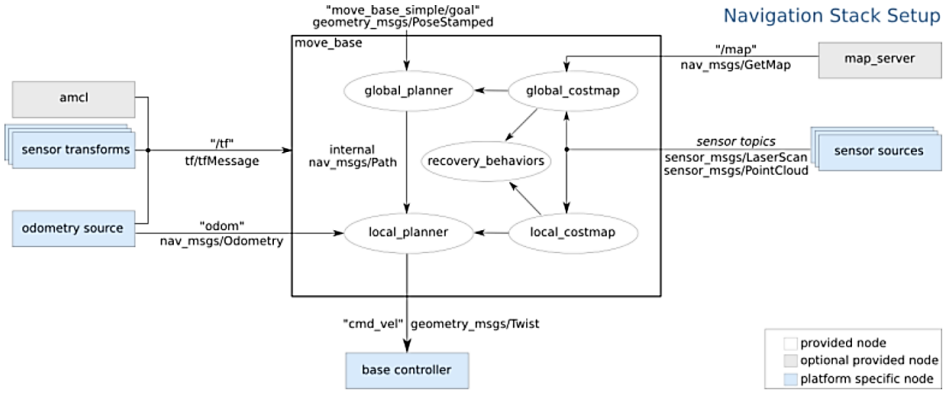
Wheeled Mobile Robot: Case 1

Slam-gmapping

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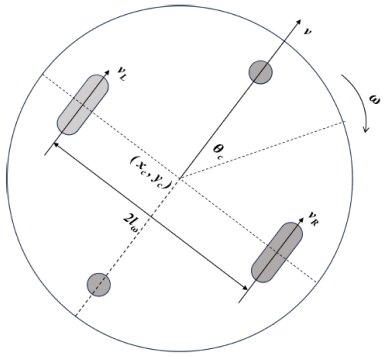


Illustration of robot chassis

Wheeled Mobile Robot: Case 2

- **Waymo**: The current global leader in fully driverless, paid robotaxi service and safety transparency, operating rider-only services at scale in Phoenix, San Francisco, Los Angeles, Austin (and more), surpassing 100M driverless miles by mid-2025.



Wheeled Mobile Robot: Case 2



- 6th-generation Waymo Driver (since Nov. 2025)
  - Sensor suite (counts & range): Cameras: 13. Lidar: 4. Radar: 6. External Audio Receivers (EARs): present (for sirens, etc.). Designed for overlapping 360° fields and visibility up to ~500 m in day/night across a wider range of weather
  - Vehicle platforms (integration) Being integrated on Zeekr (RT/Mix-derived) and **Hyundai Ioniq 5** platforms as Waymo scales manufacturing beyond the Jaguar fleet.

Wheeled Mobile Robot: Case 2

- Tesla Cybercab (production since Q2 2026)

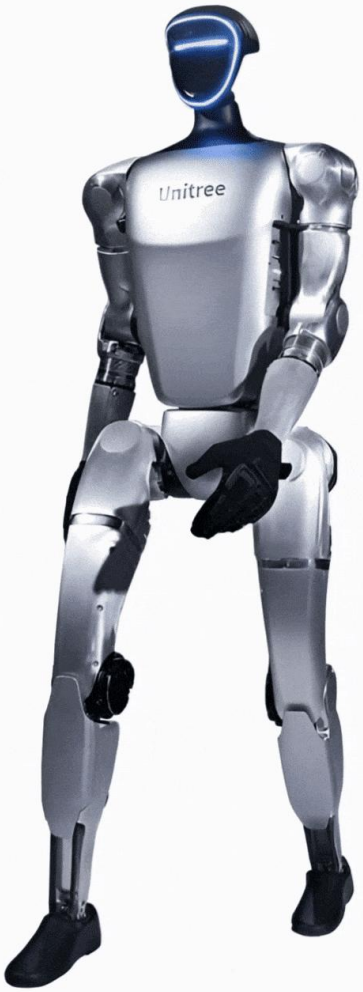
  - Cameras: 7 exterior cameras, plus 1 cabin camera for occupancy monitoring. Lidar: none. Radar: none. Cybercab follows Tesla's vision-based autonomous-driving approach, using onboard AI to interpret camera data.

  - A purpose-built, two-seat robotaxi with butterfly doors and no steering wheel, accelerator, or brake pedal. Production began at Gigafactory Texas in Q2 2026, and limited Cybercab rides are currently available in parts of Austin.



Legged Mobile Robot: Case 3

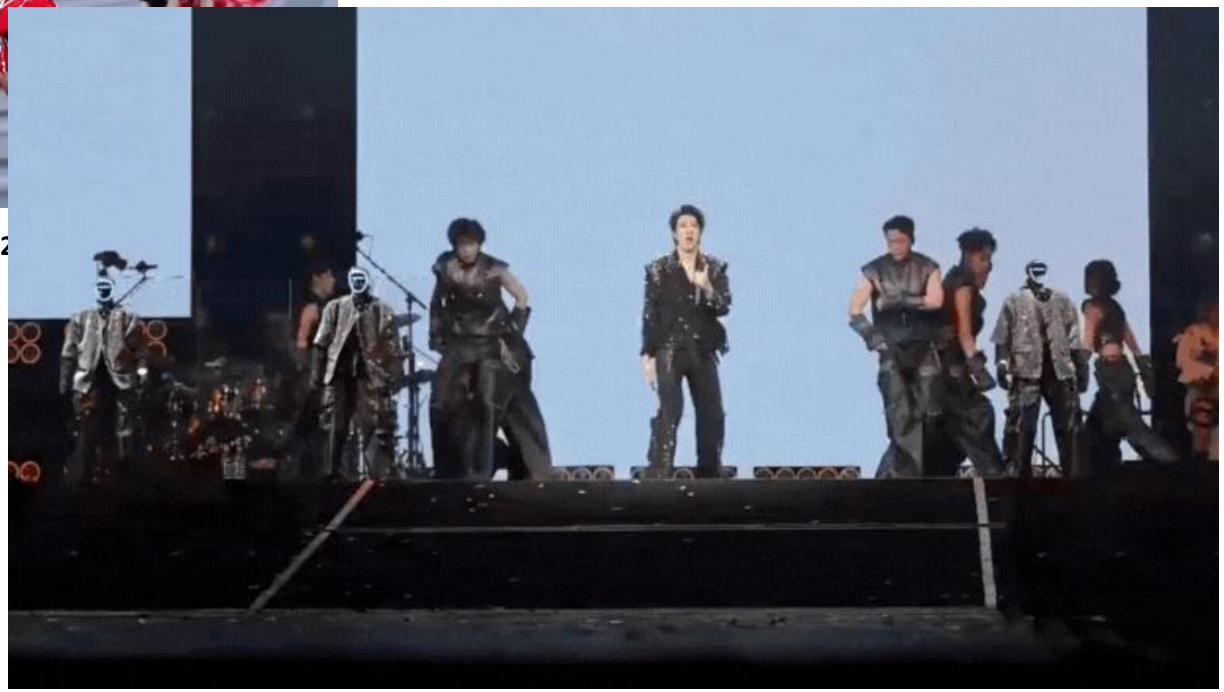
- **Unitree G1**: A compact, AI-driven humanoid designed for research, education, and light manipulation tasks, with optional dexterous hands and developer “EDU” configurations.
- Price: From US$13.5k–16k depending on configuration (G1 base vs G1 EDU), making it one of the most accessible humanoids.
- Headline capability: Up to 43 total joints in EDU configs; demonstrations include side-flips and kip-up recovery showing high agility for its size.



Legged Mobile Robot: Case 3



Unitree G1 at the beginning of 2



Unitree G1 at the end of 2025

Legged Mobile Robot: Case 3



Unitree G1 in2026



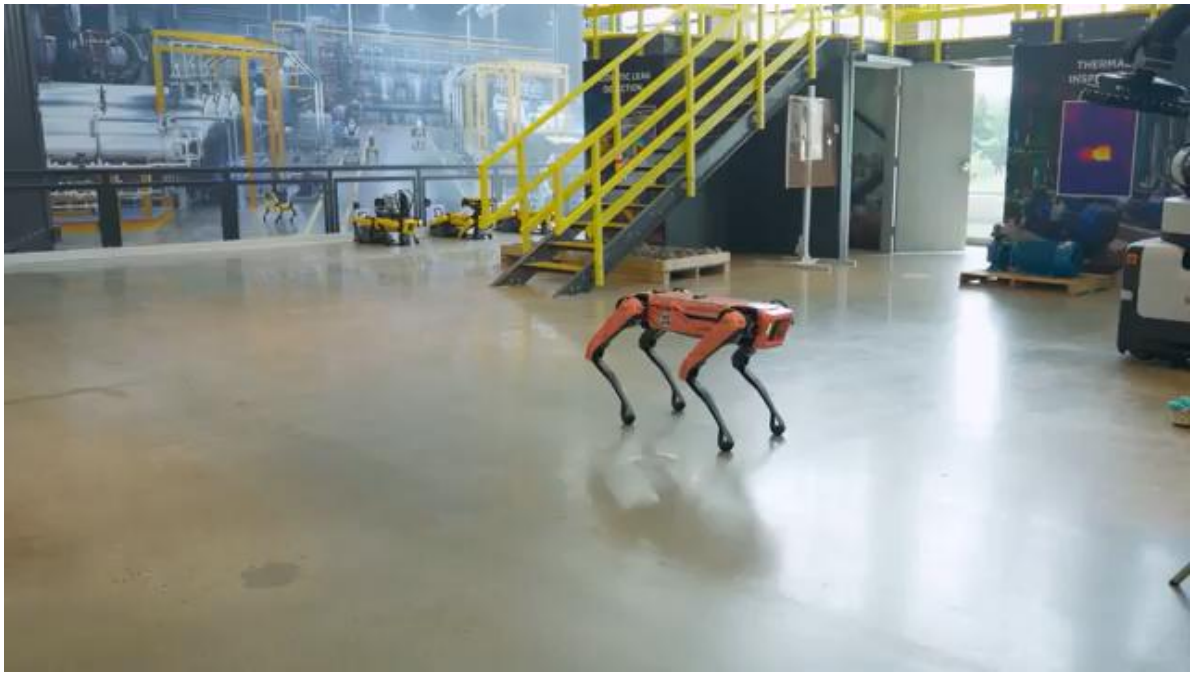
Lightning humanoid robot by Honor and a Tiangong Ultra humanoid robot compete in a 100-meter exhibition race

Legged Mobile Robot: Case 3

Category	Unitree G1	Boston Dynamics Atlas
Price	**US$13.5k–16k** (base G1; EDU is custom-quoted depending on modules)	Not sold / price not disclosed (estimated to be **US$140k**)
Availability / target use	Commercially available; research/education & light manipulation (EDU options for secondary development)	R&D platform used by BD to explore humanoid mobility & manipulation; showcased in factory pilots
Height	**~1.32 m**	**~1.50 m**
Mass	**~35 kg**	**~89 kg**
Degrees of Freedom (DoF)	**23 (base) → up to 43 (EDU)** — extra waist/wrist & 3-finger dexterous hands optional	**28 joints**
Actuation	Low-inertia internal-rotor **PMSM** motors; crossed-roller joint bearings	**Fully electric** custom actuators; high strength-to-weight components (titanium/aluminum, 3D-printed parts)
Knee torque (max)	**90 N·m (base) → 120 N·m (EDU)**	(Not publicly specified)
Arm payload (typ.)	**~2 kg (base) → ~3 kg (EDU)**	(Not publicly specified in Atlas

Legged Mobile Robot: Case 4

- **Boston Dynamics Spot**: Four-legged “robot dog” built to automate sensing, inspection, and data capture across industrial sites. Core capabilities: 360° perception, autonomous & teleop missions, self-righting and docking; carries up to 14 kg of payloads with 90 min typical runtime and robust navigation over stairs and rough terrain.



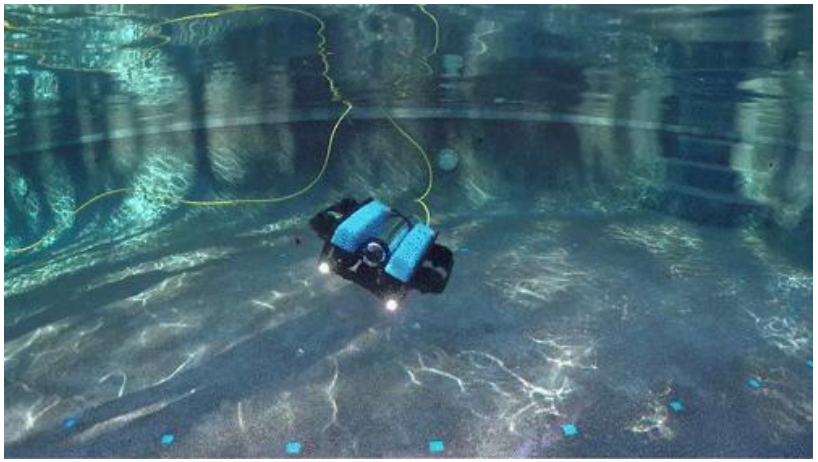
Aerial Mobile Robot: Case 5

- Holybro X500: A UAV digital twin framework for remote sensing in the Andes built using ROS, Gazebo, and ArduPilot. Environment reconstructed from satellite data;



Aquatic Mobile Robot: Case 6

- **BlueROV2**: An autonomous underwater vehicle (AUV) used for oceanographic surveys, seabed mapping, and environmental monitoring.



- Key Features: High endurance, equipped with sonar and environmental sensors, depth capability up to thousands of meters, commonly used by research institutions and naval operations.

Hybrid Mobile Robot: Case 7



- **Aqua2**: A hybrid mobile robot that can walk on land and swim underwater, using six legs for terrestrial locomotion and fin-like motions for aquatic propulsion.
- **Key Innovation**: Combines legged locomotion and swimming in one platform, enabling seamless transitions between land and water environments.

ROSE5730 Autonomous Mobile Robots

Lecture 1: Introduction

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